

4.6 Generalized Linear Models

In Chapter 3, we assumed that the response Y is quantitative, and explored the use of least squares linear regression to predict Y . Thus far in this chapter, we have instead assumed that Y is qualitative. However, we may sometimes be faced with situations in which Y is neither qualitative nor quantitative, and so neither linear regression from Chapter 3 nor the classification approaches covered in this chapter is applicable.

As a concrete example, we consider the **Bikeshare** data set. The response is **bikers**, the number of hourly users of a bike sharing program in Washington, DC. This response value is neither qualitative nor quantitative: instead, it takes on non-negative integer values, or *counts*. We will consider predicting **bikers** using the covariates **mnth** (month of the year), **hr** (hour of the day, from 0 to 23), **workingday** (an indicator variable that equals 1 if it is neither a weekend nor a holiday), **temp** (the normalized temperature, in Celsius), and **weathersit** (a qualitative variable that takes on one of four possible values: clear; misty or cloudy; light rain or light snow; or heavy rain or heavy snow.)

In the analyses that follow, we will treat **mnth**, **hr**, and **weathersit** as qualitative variables.

4.6.1 Linear Regression on the Bikeshare Data

To begin, we consider predicting **bikers** using linear regression. The results are shown in Table 4.10.

We see, for example, that a progression of weather from clear to cloudy results in, on average, 12.89 fewer bikers per hour; however, if the weather progresses further to rain or snow, then this further results in 53.60 fewer bikers per hour. Figure 4.13 displays the coefficients associated with **mnth** and the coefficients associated with **hr**. We see that bike usage is highest in the spring and fall, and lowest during the winter months. Furthermore, bike usage is greatest around rush hour (9 AM and 6 PM), and lowest overnight. Thus, at first glance, fitting a linear regression model to the **Bikeshare** data set seems to provide reasonable and intuitive results.

	Coefficient	Std. error	z-statistic	p-value
Intercept	73.60	5.13	14.34	0.00
workingday	1.27	1.78	0.71	0.48
temp	157.21	10.26	15.32	0.00
weathersit[cloudy/misty]	-12.89	1.96	-6.56	0.00
weathersit[light rain/snow]	-66.49	2.97	-22.43	0.00
weathersit[heavy rain/snow]	-109.75	76.67	-1.43	0.15

TABLE 4.10. Results for a least squares linear model fit to predict **bikers** in the **Bikeshare** data. The predictors **mnth** and **hr** are omitted from this table due to space constraints, and can be seen in Figure 4.13. For the qualitative variable **weathersit**, the baseline level corresponds to clear skies.

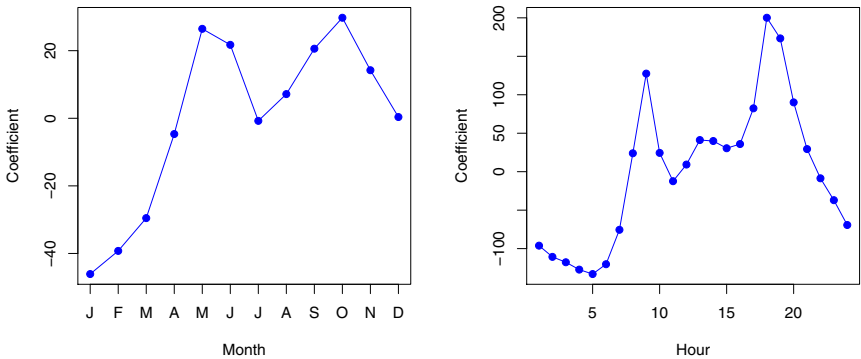


FIGURE 4.13. A least squares linear regression model was fit to predict **bikers** in the **Bikeshare** data set. Left: The coefficients associated with the month of the year. Bike usage is highest in the spring and fall, and lowest in the winter. Right: The coefficients associated with the hour of the day. Bike usage is highest during peak commute times, and lowest overnight.

But upon more careful inspection, some issues become apparent. For example, 9.6% of the fitted values in the **Bikeshare** data set are negative: that is, the linear regression model predicts a *negative* number of users during 9.6% of the hours in the data set. This calls into question our ability to perform meaningful predictions on the data, and it also raises concerns about the accuracy of the coefficient estimates, confidence intervals, and other outputs of the regression model.

Furthermore, it is reasonable to suspect that when the expected value of **bikers** is small, the variance of **bikers** should be small as well. For instance, at 2 AM during a heavy December snow storm, we expect that extremely few people will use a bike, and moreover that there should be little variance associated with the number of users during those conditions. This is borne out in the data: between 1 AM and 4 AM, in December, January, and February, when it is raining, there are 5.05 users, on average,

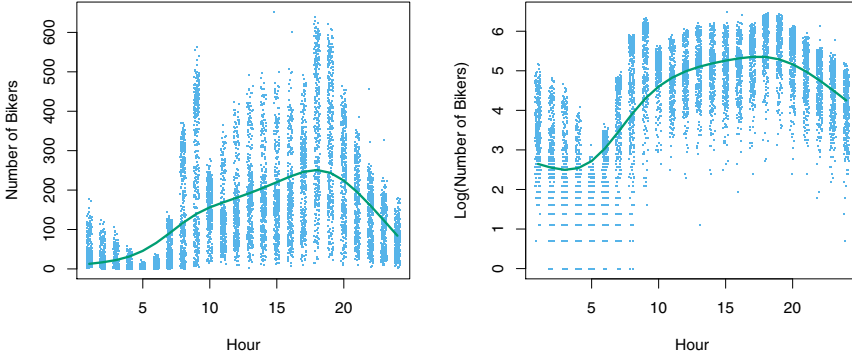


FIGURE 4.14. Left: On the **Bikeshare** dataset, the number of bikers is displayed on the y-axis, and the hour of the day is displayed on the x-axis. Jitter was applied for ease of visualization. For the most part, as the mean number of bikers increases, so does the variance in the number of bikers. A smoothing spline fit is shown in green. Right: The log of the number of bikers is now displayed on the y-axis.

with a standard deviation of 3.73. By contrast, between 7 AM and 10 AM, in April, May, and June, when skies are clear, there are 243.59 users, on average, with a standard deviation of 131.7. The mean-variance relationship is displayed in the left-hand panel of Figure 4.14. This is a major violation of the assumptions of a linear model, which state that $Y = \sum_{j=1}^p X_j \beta_j + \epsilon$, where ϵ is a mean-zero error term with variance σ^2 that is *constant*, and not a function of the covariates. Therefore, the heteroscedasticity of the data calls into question the suitability of a linear regression model.

Finally, the response **bikers** is integer-valued. But under a linear model, $Y = \beta_0 + \sum_{j=1}^p X_j \beta_j + \epsilon$, where ϵ is a continuous-valued error term. This means that in a linear model, the response Y is necessarily continuous-valued (quantitative). Thus, the integer nature of the response **bikers** suggests that a linear regression model is not entirely satisfactory for this data set.

Some of the problems that arise when fitting a linear regression model to the **Bikeshare** data can be overcome by transforming the response; for instance, we can fit the model

$$\log(Y) = \sum_{j=1}^p X_j \beta_j + \epsilon.$$

Transforming the response avoids the possibility of negative predictions, and it overcomes much of the heteroscedasticity in the untransformed data, as is shown in the right-hand panel of Figure 4.14. However, it is not quite a satisfactory solution, since predictions and inference are made in terms of the log of the response, rather than the response. This leads to challenges in interpretation, e.g. “a one-unit increase in X_j is associated with an increase in the mean of the log of Y by an amount β_j ”. Furthermore, a

log transformation of the response cannot be applied in settings where the response can take on a value of 0. Thus, while fitting a linear model to a transformation of the response may be an adequate approach for some count-valued data sets, it often leaves something to be desired. We will see in the next section that a Poisson regression model provides a much more natural and elegant approach for this task.

4.6.2 Poisson Regression on the Bikeshare Data



To overcome the inadequacies of linear regression for analyzing the **Bikeshare** data set, we will make use of an alternative approach, called *Poisson regression*. Before we can talk about Poisson regression, we must first introduce the *Poisson distribution*.

Poisson
regression
Poisson
distribution

Suppose that a random variable Y takes on nonnegative integer values, i.e. $Y \in \{0, 1, 2, \dots\}$. If Y follows the Poisson distribution, then

$$\Pr(Y = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k = 0, 1, 2, \dots \quad (4.35)$$

Here, $\lambda > 0$ is the expected value of Y , i.e. $E(Y)$. It turns out that λ also equals the variance of Y , i.e. $\lambda = E(Y) = \text{Var}(Y)$. This means that if Y follows the Poisson distribution, then the larger the mean of Y , the larger its variance. (In (4.35), the notation $k!$, pronounced “k factorial”, is defined as $k! = k \times (k - 1) \times (k - 2) \times \dots \times 3 \times 2 \times 1$.)

The Poisson distribution is typically used to model *counts*; this is a natural choice for a number of reasons, including the fact that counts, like the Poisson distribution, take on nonnegative integer values. To see how we might use the Poisson distribution in practice, let Y denote the number of users of the bike sharing program during a particular hour of the day, under a particular set of weather conditions, and during a particular month of the year. We might model Y as a Poisson distribution with mean $E(Y) = \lambda = 5$. This means that the probability of no users during this particular hour is $\Pr(Y = 0) = \frac{e^{-5} 5^0}{0!} = e^{-5} = 0.0067$ (where $0! = 1$ by convention). The probability that there is exactly one user is $\Pr(Y = 1) = \frac{e^{-5} 5^1}{1!} = 5e^{-5} = 0.034$, the probability of two users is $\Pr(Y = 2) = \frac{e^{-5} 5^2}{2!} = 0.084$, and so on.

Of course, in reality, we expect the mean number of users of the bike sharing program, $\lambda = E(Y)$, to vary as a function of the hour of the day, the month of the year, the weather conditions, and so forth. So rather than modeling the number of bikers, Y , as a Poisson distribution with a fixed mean value like $\lambda = 5$, we would like to allow the mean to vary as a function of the covariates. In particular, we consider the following model for the mean $\lambda = E(Y)$, which we now write as $\lambda(X_1, \dots, X_p)$ to emphasize that it is a function of the covariates $X_1, \dots, X_p$:

$$\log(\lambda(X_1, \dots, X_p)) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p \quad (4.36)$$

	Coefficient	Std. error	z-statistic	p-value
Intercept	4.12	0.01	683.96	0.00
workingday	0.01	0.00	7.5	0.00
temp	0.79	0.01	68.43	0.00
weathersit[cloudy/misty]	-0.08	0.00	-34.53	0.00
weathersit[light rain/snow]	-0.58	0.00	-141.91	0.00
weathersit[heavy rain/snow]	-0.93	0.17	-5.55	0.00

TABLE 4.11. *Results for a Poisson regression model fit to predict bikers in the Bikeshare data. The predictors mnth and hr are omitted from this table due to space constraints, and can be seen in Figure 4.15. For the qualitative variable weathersit, the baseline corresponds to clear skies.*

or equivalently

$$\lambda(X_1, \dots, X_p) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}. \tag{4.37}$$

Here, $\beta_0, \beta_1, \dots, \beta_p$ are parameters to be estimated. Together, (4.35) and (4.36) define the Poisson regression model. Notice that in (4.36), we take the $\log$ of $\lambda(X_1, \dots, X_p)$ to be linear in $X_1, \dots, X_p$, rather than having $\lambda(X_1, \dots, X_p)$ itself be linear in $X_1, \dots, X_p$, in order to ensure that $\lambda(X_1, \dots, X_p)$ takes on nonnegative values for all values of the covariates.

To estimate the coefficients $\beta_0, \beta_1, \dots, \beta_p$, we use the same maximum likelihood approach that we adopted for logistic regression in Section 4.3.2. Specifically, given n independent observations from the Poisson regression model, the likelihood takes the form

$$\ell(\beta_0, \beta_1, \dots, \beta_p) = \prod_{i=1}^n \frac{e^{-\lambda(x_i)} \lambda(x_i)^{y_i}}{y_i!}, \tag{4.38}$$

where $\lambda(x_i) = e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}}$, due to (4.37). We estimate the coefficients that maximize the likelihood $\ell(\beta_0, \beta_1, \dots, \beta_p)$, i.e. that make the observed data as likely as possible.

We now fit a Poisson regression model to the Bikeshare data set. The results are shown in Table 4.11 and Figure 4.15. Qualitatively, the results are similar to those from linear regression in Section 4.6.1. We again see that bike usage is highest in the spring and fall and during rush hour, and lowest during the winter and in the early morning hours. Moreover, bike usage increases as the temperature increases, and decreases as the weather worsens. Interestingly, the coefficient associated with workingday is statistically significant under the Poisson regression model, but not under the linear regression model.

Some important distinctions between the Poisson regression model and the linear regression model are as follows:

- *Interpretation:* To interpret the coefficients in the Poisson regression model, we must pay close attention to (4.37), which states that an increase in X_j by one unit is associated with a change in $E(Y) = \lambda$ by a factor of $\exp(\beta_j)$. For example, a change in weather from clear

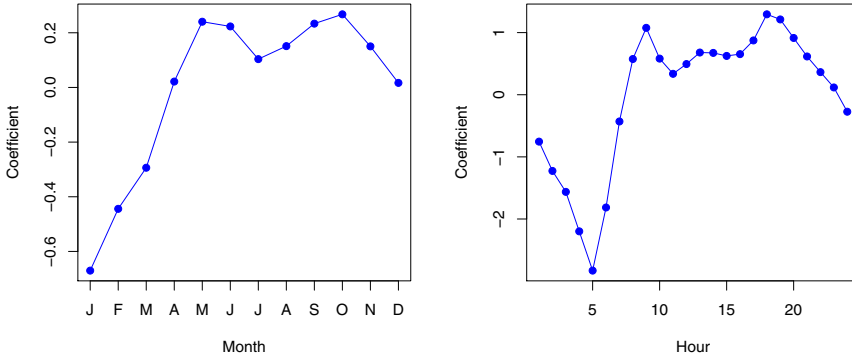


FIGURE 4.15. A Poisson regression model was fit to predict bikers in the Bikeshare data set. Left: The coefficients associated with the month of the year. Bike usage is highest in the spring and fall, and lowest in the winter. Right: The coefficients associated with the hour of the day. Bike usage is highest during peak commute times, and lowest overnight.

to cloudy skies is associated with a change in mean bike usage by a factor of $\exp(-0.08) = 0.923$, i.e. on average, only 92.3% as many people will use bikes when it is cloudy relative to when it is clear. If the weather worsens further and it begins to rain, then the mean bike usage will further change by a factor of $\exp(-0.5) = 0.607$, i.e. on average only 60.7% as many people will use bikes when it is rainy relative to when it is cloudy.

- *Mean-variance relationship:* As mentioned earlier, under the Poisson model, $\lambda = E(Y) = \text{Var}(Y)$. Thus, by modeling bike usage with a Poisson regression, we implicitly assume that mean bike usage in a given hour equals the variance of bike usage during that hour. By contrast, under a linear regression model, the variance of bike usage always takes on a constant value. Recall from Figure 4.14 that in the Bikeshare data, when biking conditions are favorable, both the mean *and* the variance in bike usage are much higher than when conditions are unfavorable. Thus, the Poisson regression model is able to handle the mean-variance relationship seen in the Bikeshare data in a way that the linear regression model is not.⁵
- *nonnegative fitted values:* There are no negative predictions using the Poisson regression model. This is because the Poisson model itself only allows for nonnegative values; see (4.35). By contrast, when we

overdispersion

⁵In fact, the variance in the Bikeshare data appears to be much higher than the mean, a situation referred to as *overdispersion*. This causes the Z-values to be inflated in Table 4.11. A more careful analysis should account for this overdispersion to obtain more accurate Z-values, and there are a variety of methods for doing this. But they are beyond the scope of this book.

fit a linear regression model to the **Bikeshare** data set, almost 10% of the predictions were negative.

4.7.7 Poisson Regression

Finally, we fit a Poisson regression model to the **Bikeshare** data set, which measures the number of bike rentals (**bikers**) per hour in Washington, DC. The data can be found in the **ISLR2** library.


```
> attach(Bikeshare)
> dim(Bikeshare)
[1] 8645 15
> names(Bikeshare)
[1] "season"      "mnth"        "day"         "hr"
[5] "holiday"     "weekday"     "workingday"  "weathersit"
[9] "temp"        "atemp"       "hum"         "windspeed"
[13] "casual"      "registered" "bikers"
```

We begin by fitting a least squares linear regression model to the data.

```
> mod.lm <- lm(
  bikers ~ mnth + hr + workingday + temp + weathersit,
  data = Bikeshare
)
> summary(mod.lm)
Call:
lm(formula = bikers ~ mnth + hr + workingday + temp +
    weathersit, data = Bikeshare)

Residuals:
    Min       1Q   Median       3Q      Max
-299.00  -45.70   -6.23   41.08  425.29

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -68.632     5.307  -12.932 < 2e-16 ***
mnthFeb         6.845     4.287    1.597 0.110398
mnthMarch      16.551     4.301    3.848 0.000120 ***
mnthApril      41.425     4.972    8.331 < 2e-16 ***
mnthMay        72.557     5.641   12.862 < 2e-16 ***
```

Due to space constraints, we truncate the output of `summary(mod.lm)`. In `mod.lm`, the first level of `hr` (0) and `mnth` (Jan) are treated as the baseline values, and so no coefficient estimates are provided for them: implicitly, their coefficient estimates are zero, and all other levels are measured relative to these baselines. For example, the Feb coefficient of 6.845 signifies that, holding all other variables constant, there are on average about 7 more riders in February than in January. Similarly there are about 16.5 more riders in March than in January.

The results seen in Section 4.6.1 used a slightly different coding of the variables `hr` and `mnth`, as follows:

```
> contrasts(Bikeshare$hr) = contr.sum(24)
> contrasts(Bikeshare$mnth) = contr.sum(12)
> mod.lm2 <- lm(
  bikers ~ mnth + hr + workingday + temp + weathersit,
  data = Bikeshare
)
> summary(mod.lm2)
Call:
lm(formula = bikers ~ mnth + hr + workingday + temp +
    weathersit, data = Bikeshare)
```

```

Residuals:
    Min       1Q   Median       3Q      Max
-299.00  -45.70   -6.23   41.08  425.29

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    73.597     5.132   14.340 < 2e-16 ***
mnth1         -46.087     4.086  -11.281 < 2e-16 ***
mnt2          -39.242     3.539  -11.088 < 2e-16 ***
mnt3          -29.536     3.155   -9.361 < 2e-16 ***
mnt4           -4.662     2.741   -1.701  0.08895 .

```

What is the difference between the two codings? In `mod.lm2`, a coefficient estimate is reported for all but the last level of `hr` and `mnt`. Importantly, in `mod.lm2`, the coefficient estimate for the last level of `mnt` is not zero: instead, it equals the *negative of the sum of the coefficient estimates for all of the other levels*. Similarly, in `mod.lm2`, the coefficient estimate for the last level of `hr` is the negative of the sum of the coefficient estimates for all of the other levels. This means that the coefficients of `hr` and `mnt` in `mod.lm2` will always sum to zero, and can be interpreted as the difference from the mean level. For example, the coefficient for January of -46.087 indicates that, holding all other variables constant, there are typically 46 fewer riders in January relative to the yearly average.

It is important to realize that the choice of coding really does not matter, provided that we interpret the model output correctly in light of the coding used. For example, we see that the predictions from the linear model are the same regardless of coding:

```

> sum((predict(mod.lm) - predict(mod.lm2))^2)
[1] 1.426e-18

```

The sum of squared differences is zero. We can also see this using the `all.equal()` function:

`all.equal()`

```

> all.equal(predict(mod.lm), predict(mod.lm2))

```

To reproduce the left-hand side of Figure 4.13, we must first obtain the coefficient estimates associated with `mnt`. The coefficients for January through November can be obtained directly from the `mod.lm2` object. The coefficient for December must be explicitly computed as the negative sum of all the other months.

```

> coef.months <- c(coef(mod.lm2)[2:12],
  -sum(coef(mod.lm2)[2:12]))

```

To make the plot, we manually label the x -axis with the names of the months.

```

> plot(coef.months, xlab = "Month", ylab = "Coefficient",
  xaxt = "n", col = "blue", pch = 19, type = "o")

```

```
> axis(side = 1, at = 1:12, labels = c("J", "F", "M", "A",
  "M", "J", "J", "A", "S", "O", "N", "D"))
```

Reproducing the right-hand side of Figure 4.13 follows a similar process.

```
> coef.hours <- c(coef(mod.lm2)[13:35],
  -sum(coef(mod.lm2)[13:35]))
> plot(coef.hours, xlab = "Hour", ylab = "Coefficient",
  col = "blue", pch = 19, type = "o")
```

Now, we consider instead fitting a Poisson regression model to the *Bikeshare* data. Very little changes, except that we now use the function `glm()` with the argument `family = poisson` to specify that we wish to fit a Poisson regression model:

```
> mod.pois <- glm(
  bikers ~ mnth + hr + workingday + temp + weathersit,
  data = Bikeshare, family = poisson
)
> summary(mod.pois)
Call:
glm(formula = bikers ~ mnth + hr + workingday + temp +
  weathersit, family = poisson, data = Bikeshare)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-20.7574	-3.3441	-0.6549	2.6999	21.9628

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.118245	0.006021	683.964	< 2e-16 ***
mnth1	-0.670170	0.005907	-113.445	< 2e-16 ***
mnth2	-0.444124	0.004860	-91.379	< 2e-16 ***
mnth3	-0.293733	0.004144	-70.886	< 2e-16 ***
mnth4	0.021523	0.003125	6.888	5.66e-12 ***

We can plot the coefficients associated with `mnth` and `hr`, in order to reproduce Figure 4.15:

```
> coef.mnth <- c(coef(mod.pois)[2:12],
  -sum(coef(mod.pois)[2:12]))
> plot(coef.mnth, xlab = "Month", ylab = "Coefficient",
  xaxt = "n", col = "blue", pch = 19, type = "o")
> axis(side = 1, at = 1:12, labels = c("J", "F", "M", "A", "M",
  "J", "J", "A", "S", "O", "N", "D"))
> coef.hours <- c(coef(mod.pois)[13:35],
  -sum(coef(mod.pois)[13:35]))
> plot(coef.hours, xlab = "Hour", ylab = "Coefficient",
  col = "blue", pch = 19, type = "o")
```

We can once again use the `predict()` function to obtain the fitted values (predictions) from this Poisson regression model. However, we must use the argument `type = "response"` to specify that we want *R* to output $\exp(\hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p)$ rather than $\hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$, which it will output by default.

```
> plot(predict(mod.lm2), predict(mod.pois, type = "response"))  
> abline(0, 1, col = 2, lwd = 3)
```

The predictions from the Poisson regression model are correlated with those from the linear model; however, the former are non-negative. As a result the Poisson regression predictions tend to be larger than those from the linear model for either very low or very high levels of ridership.

In this section, we used the `glm()` function with the argument `family = poisson` in order to perform Poisson regression. Earlier in this lab we used the `glm()` function with `family = binomial` to perform logistic regression. Other choices for the `family` argument can be used to fit other types of GLMs. For instance, `family = Gamma` fits a gamma regression model.